

SIMATS ENGINEERING

ASSESSMENT TOOL 1 :

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Concept Mapping COURSE NAME:

Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

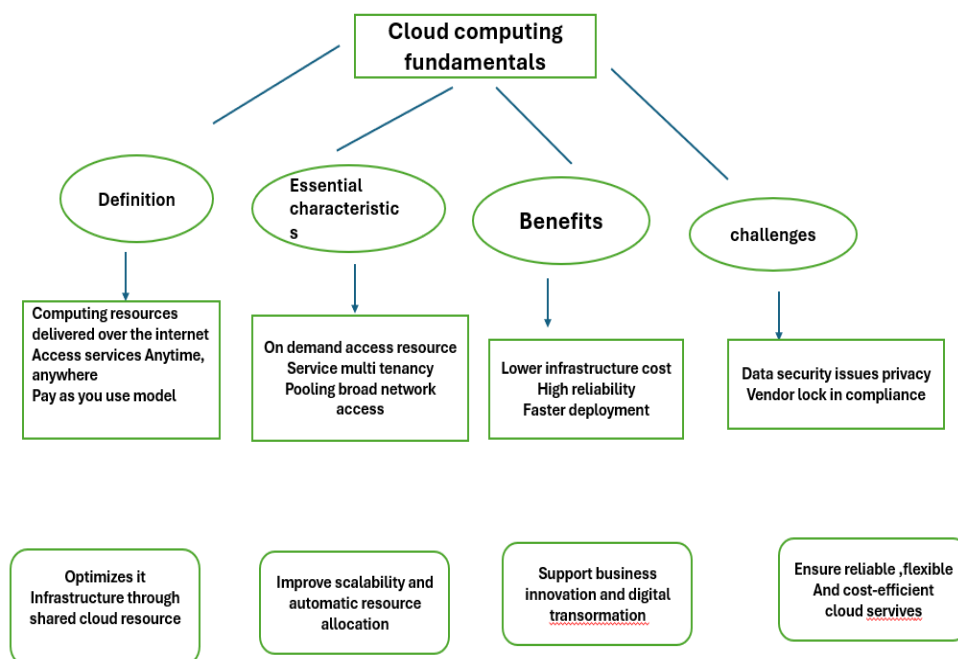
COURSE OUTCOME COVERED CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20% Dave's Taxonomy: Imitation (Level 1)

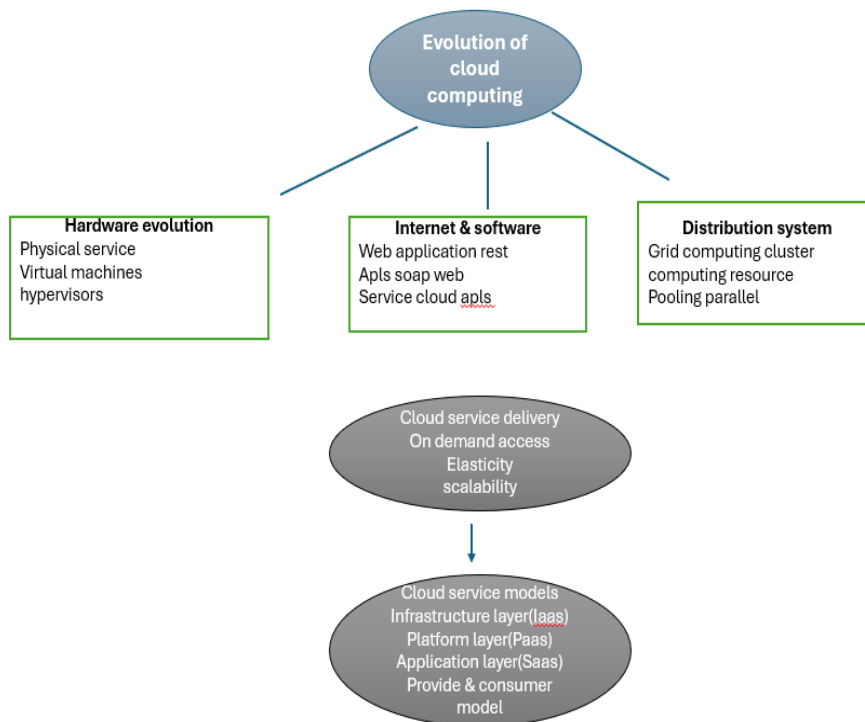
Question Count: 4

Total Marks: 50

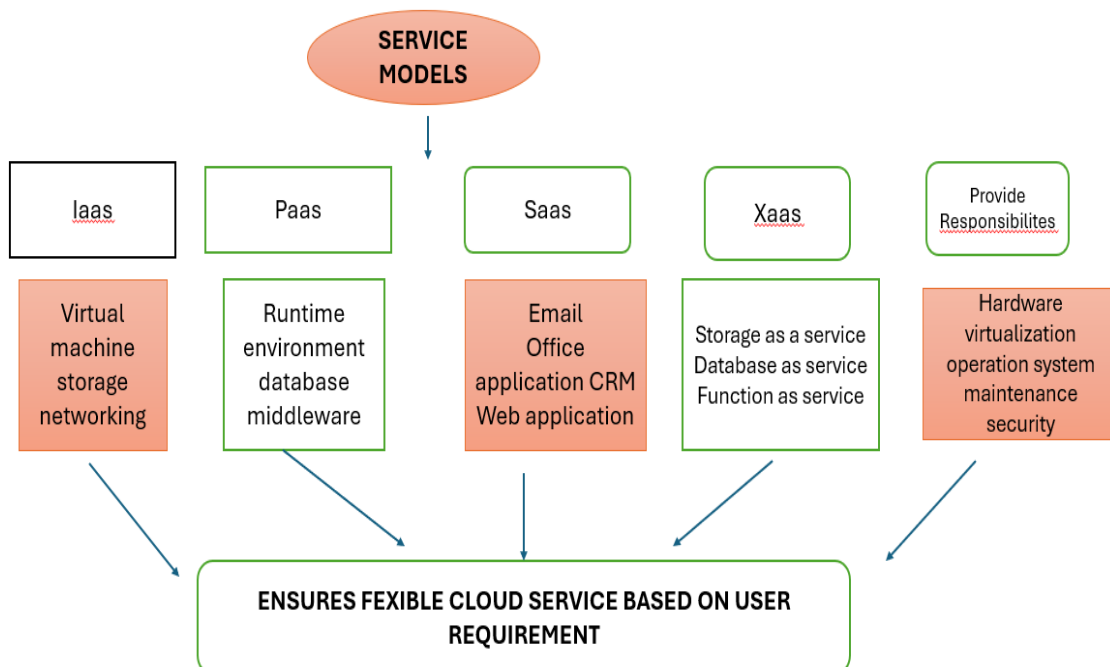
Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



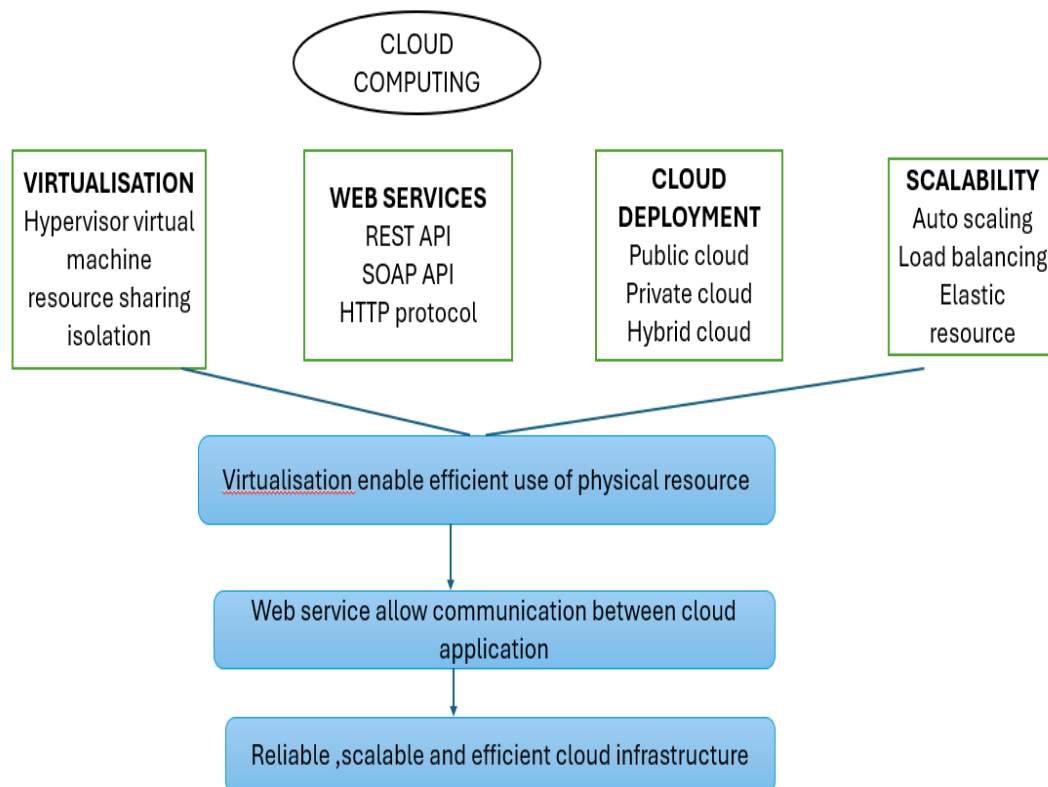
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



3. Concept Map 3: Service Models. Map laaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



1). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why? The concept map effectively represents cloud computing by showing the connection between cloud characteristics, service models, deployment models, and applications. The use of clear categories and arrows makes the relationships easy to understand. Grouping related concepts together helps explain how cloud computing works as a complete system.

2). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent? Scalability allows the system to handle increasing workloads, while elasticity automatically adds or removes resources based on demand. Resource management ensures efficient allocation of computing, storage, and network resources. Without these features, the system could experience slow performance, resource wastage, higher costs, and difficulty handling large numbers of users.

3). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system? APIs and web services enable communication between cloud applications and users. Cloud services such as IaaS, PaaS, and SaaS provide infrastructure, development platforms, and software solutions. These services improve system functionality by enabling easy access, faster deployment, reduced maintenance, and better resource utilization.

4). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience? The relationships between cloud infrastructure, virtualization, storage, networking, and service models ensure that resources work together efficiently. These interactions improve reliability through backup and fault tolerance, enhance performance through load balancing and resource sharing, and provide users with secure, fast, and continuous access to cloud services.

5). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it? Developing the concept map improved my understanding of how different cloud computing components are connected and how they work together. It helped me visualize the architecture more clearly. If I redesigned the map, I would include more real-world examples, security features, and detailed connections between cloud services to make it more informative and easier to understand.